

Zhi Chen

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EDUCATION

- **Stony Brook University** Stony Brook, NY
Doctor of Philosophy - Physics; *Aug 2023 - Expected May 2027*
- **Stony Brook University** Stony Brook, NY
Master of Arts - Physics *Aug 2021 - May 2023*
Thesis: Sensitivity of He Flames in X-Ray Bursts to Nuclear Physics
- **Stony Brook University** Stony Brook, NY
Bachelor of Science - Physics; Summa Cum Laude *Aug 2017 - May 2021*

RESEARCH EXPERIENCE

- **PhD Thesis Research** Aug 2023 - Present
Graduate Student Research
 - Extended the finite-volume Godunov scheme in *Castro* to axisymmetric spherical geometry for multidimensional simulations of Helium deflagration in Type I X-ray bursts, investigating the influence of rotation on the burst dynamics.
 - Implemented nuclear statistical equilibrium (NSE) evolution algorithm in *Microphysics* for efficient high-temperature reaction network integration in large-scale astrophysical reactive flow simulations.
 - Contributed to multiple open-source astronomy softwares, including *Pynucastro*, *Microphysics*, *Castro*, and *pyro2*, with over 300 merged pull requests across these projects.
- **Master's Thesis Research** Jan 2022 - Aug 2023
Graduate Student Research
 - Investigated the effects of nuclear reaction networks, electron screening routines, and hydro-reaction coupling methods on lateral Helium flame propagation in Type I X-ray bursts using *Castro*, an adaptive mesh, compressible hydrodynamics simulation code.
- **SULI Program at Brookhaven National Lab** Jun 2021 - Jan 2022
Summer Internship + Continued Research
 - Developed a Python *simulation code* that simulated the theoretical observational signal through two-photon interferometry given a set of telescopes and star pairs. With the simulated signal, a Markov Chain Monte Carlo algorithm was used to estimate the precision of the relative separation between the star pair.
- **SULI Program at Brookhaven National Lab** Jun 2020 - Aug 2020
Summer Internship
 - Developed additional features for *imcurio*, a Python simulation code for 21 cm intensity mapping observations that simulates a fixed set of visibilities for a fixed telescope and fixed sky.

PUBLICATIONS

- **Chen, Z.**, Bhargava, K., Johnson, E., Lopes, R., Rasmussen, M., Smith Clark, A., Zingale, M., Boyd, B., Eiden, K., Glosser, S., & Willcox, D. (2026): *pynucastro 3: A community library for nuclear astrophysics*. DOI: 10.48550/arXiv.2608.17049
- AMReX-Astro Microphysics Team, Bhargava, K., Bishop, A., **Chen, Z.**, Fan, D., Fields, C., Jacobs, A. M., Johnson, E. T., Katz, M., Krumholz, M., Malone, C., Nonaka, A., Sharda, P., Smith Clark, A., Timmes, F., Wibking, B., Willcox, D. E., & Zingale, M. (2026): *AMReX-Astrophysics Microphysics: A set of microphysics routines for astrophysical simulation codes based on the AMReX library*. DOI: 10.21105/joss.10422
- Zingale, M., Bhargava, K., Brady, R., **Chen, Z.**, Guichandut, S., Johnson, E. T., Katz, M., & Smith Clark, A. (2025): *The Challenges of Modeling Astrophysical Reacting Flows*. DOI: 10.1088/1742-6596/2997/1/012007
- Zingale, M., **Chen, Z.**, Johnson, E. T., Katz, M. P., & Smith Clark, A. (2024): *Strong Coupling of Hydrodynamics and Reactions in Nuclear Statistical Equilibrium for Modeling Convection in Massive Stars*. DOI: 10.3847/1538-4357/ad8a66
- Zingale, M., **Chen, Z.**, Rasmussen, M., Polin, A., Katz, M., Smith Clark, A., & Johnson, E. T. (2024): *Sensitivity of Simulations of Double-detonation Type Ia Supernovae to Integration Methodology*. DOI: 10.3847/1538-4357/ad3441
- Clark, A., Johnson, E., **Chen, Z.**, Eiden, K., Zingale, M., Boyd, B., Johnson, P., & DaCosta, L.. (2024): *pynucastro 2.1: an update on the development of a python library for nuclear astrophysics*. DOI: 10.1088/1742-6596/2742/1/012003
- **Chen, Z.**, Johnson, E. T., Katz, M., Smith Clark, A., Boyd, B., & Zingale, M. (2024): *A Framework for Exploring Nuclear Physics Sensitivity in Numerical Simulations*. DOI: 10.1088/1742-6596/2742/1/012021
- **Chen, Z.**, Zingale, M., & Eiden, K. (2023): *Sensitivity of He Flames in X-Ray Bursts to Nuclear Physics*. DOI: 10.3847/1538-4357/acec72
- Smith, A. I., Johnson, E. T., **Chen, Z.**, Eiden, K., Willcox, D. E., Boyd, B., Cao, L., DeGrendele, C. J., & Zingale, M. (2023): *pynucastro: A Python Library for Nuclear Astrophysics*. DOI: 10.3847/1538-4357/acbaff
- **Chen, Z.**, Nomerotski, A., Slosar, A., Stankus, P., & Vintskevich, S. (2023): *Astrometry in two-photon interferometry using an Earth rotation fringe scan*. DOI: 10.1103/PhysRevD.107.023015
- Keach, M., Bellavia, S., **Chen, Z.**, Crawford, J., Dolzhenko, D., Figueroa, E., Mueninghoff, A., Nomerotski, A., Schiff, J., Simovitch, R., Slosar, A., Stankus, P., & Vintskevich, S. (2022): *Increasing baselines and precision of optical interferometers using two-photon interference effects*. DOI: 10.1117/12.2632122

CONFERENCES AND MEETINGS

- **Contributed Talk** at CeNAM Frontiers in Nuclear Astrophysics Meeting, 2026: *Simulating Lateral Flame Propagation in Type I X-ray Bursts*
- **Poster** Presentation at OLCF User Meeting, 2025: *Simulating X-ray Burst Flame Propagation: From Parallel Plane to Full Star*
- **Poster** Presentation at Rise.Time Meeting, 2024: *Adaptive Nuclear Statistical Equilibrium for Type Ia SN*
- **Poster** Presentation at Astronum Meeting, 2023: *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*
- **Contributed Talk** at the main CeNAM Frontiers in Nuclear Astrophysics Meeting, 2023: *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*
- SULI Internship Presentation at Brookhaven National Laboratory, Summer 2021: *Two-Photon Interferometry Simulations*
- SULI Internship Presentation at Brookhaven National Laboratory, Summer 2020: *Hydrogen Intensity Mapping Simulator: interpolation improvements and radio point source compatibility with application to GaLactic and Extragalactic All-Sky MWA Survey (GLEAM) dataset*

HONORS, GRANTS, AWARDS, AND ACHIEVEMENTS

- Gerry Brown Award (2025): Awarded for excellent research and travel funding
- Lourie Summer Fellowship (2024)
- Di-Tian Prize (2023): Awarded for excellent research and travel funding
- Stony Brook University, Academic Dean's List, 2017-2021
- NYS School Academic Excellence Scholarship

SKILLS SUMMARY

- Languages: Python, C++, Bash
- Tools: L^AT_EX, NumPy, SciPy, Sphinx
- Visualization: Matplotlib, yt
- Version Control: git, github
- Platforms: Linux
- HPC System: NESRC Perlmutter, OLCF Frontiers, ALCF Polaris